

Homology Class Asymmetry in the Loxodromic Twist Spectrum of Compact Hyperbolic 3-Manifolds

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Abstract

We introduce a homology-resolved spectral floor invariant for compact hyperbolic 3-manifolds with $H_1(M, \mathbb{Z}) = \mathbb{Z}/5$. The abelianization map partitions loxodromic elements into five homology classes; the spectral floor $\phi_{\text{floor}}^{(k)}(L)$ is the minimum folded twist angle among loxodromics of word length $\leq L$ in class k . A complete census of `m006` (Orientable-ClosedCensus[43], vol = 2.029, $H_1 = \mathbb{Z}/5$) to word length 12 (1,062,802 genuine loxodromics) shows that all five floors decay exponentially with word length, with class-dependent rates forming a paired hierarchy: $c_1 = c_4 \approx 0.985$, $c_2 = c_3 \approx 0.401$, $c_0 \approx 0.750$, confirmed by AIC model selection against power-law alternatives ($\Delta\text{AIC} > 40$ for classes 0, 1, 4). At length 12 the floors satisfy an approximate 4:2:1 ratio. The same pairing structure appears for the Meyerhoff manifold `m003` (also $H_1 = \mathbb{Z}/5$), suggesting the hierarchy is a property of $\mathbb{Z}/5$ torsion rather than of either specific manifold. We conjecture that the pairing $c_k = c_{5-k}$ reflects complex conjugation in the invariant trace field.

1 Introduction

The complex length spectrum $\{\ell(\gamma) + i\phi(\gamma)\}_\gamma$ of a compact hyperbolic 3-manifold is a fundamental geometric invariant. For manifolds with non-trivial first homology, the abelianization $\pi_1(M) \rightarrow H_1(M, \mathbb{Z})$ partitions geodesics into homology classes. A natural question is whether the twist angle distribution depends on the homology class.

We study two compact arithmetic hyperbolic 3-manifolds with $H_1 = \mathbb{Z}/5$: the Meyerhoff manifold $M_1 = \text{m003}$ (OrientableClosedCensus[1], vol = 0.9814) and $M_2 = \text{m006}$ (Orientable-ClosedCensus[43], vol = 2.029) [1, 2]. Both arise in the Hyperbolic Flavor Geometry programme [3] as geometric models for Standard Model lepton and quark mixing.

We define the *spectral floor*

$$\phi_{\text{floor}}^{(k)}(L) = \min\{\phi_{\text{fold}}(\gamma) : [\gamma] = k, |w(\gamma)| \leq L\}, \quad (1)$$

where $\phi_{\text{fold}} = \min(|\phi|, \pi - |\phi|) \in [0, \pi/2]$ and $[w] = \sum_i \varepsilon(g_i) \bmod 5$ with $\varepsilon(g) = +1$ (lowercase) or -1 (uppercase). We show that this invariant reveals a class-dependent hierarchy absent from the unpartitioned spectrum.

2 Setup and census

Twist angles $\phi(\gamma) = \arg \lambda(\gamma)$ are computed via SnapPy's `G.SL2C(word)` at 150-bit precision [10]. We enumerate all reduced words up to length L over generators $\{a, b, A, B\}$ ($A = a^{-1}$, $B = b^{-1}$), retain genuine loxodromics ($|\text{Tr}(\rho(w))| > 2.01$, equivalently $|\lambda| > 1.01$), and record $\phi_{\text{fold}}(w)$ per homology class $[w] \in \mathbb{Z}/5$. At length 12: 1,062,802 genuine loxodromics from 1,062,880 words evaluated in 84.6 seconds on a standard laptop. Near- π twists are verified via $|\text{Im}(\lambda)/\text{Re}(\lambda)|$ directly.

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3 Results

3.1 The 4:2:1 hierarchy at length 12

Table 1 gives the length-12 spectral floors per homology class for m006. The five classes split into three groups with floors in ratio approximately 4:2:1:

Table 1: Spectral floors per homology class at word length ≤ 12 for m006 (1,062,802 loxodromics). The ratio $\phi^{(0)} : \phi^{(2,3)} : \phi^{(1,4)} \approx 4:2:1$ is stable across lengths ≥ 10 .

Class k	$\phi_{\text{floor}}^{(k)}(12)$ ($^\circ$)	Word	Length
0	0.02300	aaaaaaaaabAb	12
1	0.00530	aaaaaababAAA	12
2	0.01060	aaababaaabab	12
3	0.01060	AAABABAAABAB	12
4	0.00530	aaaBAAABAAAA	12
Ratio $\phi^{(0)} : \phi^{(2,3)} : \phi^{(1,4)}$		4.3 : 2 : 1	

The pairing $\phi^{(1)} = \phi^{(4)}$ and $\phi^{(2)} = \phi^{(3)}$ reflects the involution $k \mapsto -k \pmod{5}$ on $H_1 = \mathbb{Z}/5$.

3.2 Exponential decay with class-dependent rates

Over the range $L = 1, \dots, 12$, each class exhibits log-linear decay:

$$\phi_{\text{floor}}^{(k)}(L) \sim A_k e^{-c_k L}. \quad (2)$$

Table 2 gives the fitted rates c_k with bootstrap confidence intervals and AIC model comparison.

Table 2: Exponential decay rates c_k from log-linear fits to $\phi_{\text{floor}}^{(k)}(L)$, $L = 1, \dots, 12$. Bootstrap 95% CIs (2000 resamples). $\Delta\text{AIC} = \text{AIC}(\text{exp}) - \text{AIC}(\text{pow})$; negative values favour exponential over power-law. The CIs of c_1, c_4 overlap completely; likewise c_2, c_3 .

Class k	c_k	95% CI	ΔAIC
0	0.750	[0.434, 0.944]	-33.6
1	0.985	[0.541, 1.376]	-49.0
2	0.401	[0.080, 0.645]	-4.0
3	0.401	[0.079, 0.651]	-4.0
4	0.985	[0.585, 1.431]	-49.0

The exponential model is preferred for all five classes. The pairing $c_1 = c_4 \approx 0.985$ (fastest decay) and $c_2 = c_3 \approx 0.401$ (slowest) with class 0 intermediate ($c_0 \approx 0.750$) is the principal structural result.

3.3 Universality: Meyerhoff manifold

Table 3 shows that m003 (Meyerhoff manifold) exhibits the *same* pairing at length 12, with ratio $\phi^{(2,3)}/\phi^{(1,4)} \approx 2.17$.

The appearance of the same pairing in two arithmetically distinct manifolds sharing only $H_1 = \mathbb{Z}/5$ suggests the hierarchy is a property of the $\mathbb{Z}/5$ torsion structure.

Table 3: Length-12 spectral floors for both manifolds. Both exhibit the pairing $\phi^{(1,4)} < \phi^{(0)} < \phi^{(2,3)}$ consistent with the involution $k \mapsto -k \pmod{5}$.

Class k	m003		m006	
	floor ($^\circ$)	Word	floor ($^\circ$)	Word
0	0.04373	aaaabbbbaabaB	0.02300	aaaaaaaaabAb
1	0.04319	aaBAbAAbAABB	0.00530	aaaaaababAAA
2	0.09387	aabAAbAAABB	0.01060	aaababaaabab
3	0.09387	aaaaBabbAAB	0.01060	AAABABAAABAB
4	0.04319	aabbAAbaBaaB	0.00530	aaaBAAABAAAA
$\phi^{(2,3)}/\phi^{(1,4)}$		2.17	2.00	

3.4 The twist-suppressed geodesic

The class-4 floor is set from length 6 by word AAABAB:

$$\begin{aligned}
 |\lambda(\text{AAABAB})| &= 3.730, & \phi(\text{AAABAB}) &= 179.995^\circ, & \phi_{\text{fold}} &= 0.0053^\circ, \\
 \ell(\text{AAABAB}) &= 2.633, & |\text{Tr}| &= 3.998, & [\text{AAABAB}] &= 4 \in \mathbb{Z}/5.
 \end{aligned}$$

The element is a near-pure translation ($\lambda \approx -3.730$, nearly real negative) in a non-identity homology class, demonstrating that twist suppression and non-trivial homology class are geometrically compatible.

4 Discussion

Proposition 1. *For any compact hyperbolic 3-manifold M with $H_1 = \mathbb{Z}/5$ and any class $k \in \mathbb{Z}/5$, $\phi_{\text{floor}}^{(k)}(L) \rightarrow 0$ as $L \rightarrow \infty$.*

Proof. By the Ambient Prime Geodesic Theorem [6] and equidistribution of homology classes [8], the count of class- k geodesics with $\phi_{\text{fold}} < \epsilon$ and length $\leq T$ grows as $(4\epsilon/\pi) \cdot e^{2T}/(10T)$, which diverges for any fixed $\epsilon > 0$. $\square$

Remark 1. Proposition 1 guarantees convergence to zero but not the rate. The class-dependent rates c_k (Table 2) are the new content of this paper. The exponential fits are empirical over $L = 1, \dots, 12$; no claim is made regarding the asymptotic rate as $L \rightarrow \infty$.

The pairing $c_1 = c_4$, $c_2 = c_3$ reflects the involution $k \mapsto -k \pmod{5}$ on $\mathbb{Z}/5$, since $1 + 4 \equiv 2 + 3 \equiv 0$. We propose that this is arithmetic in origin:

Conjecture 1. *The rate pairing $c_k = c_{5-k}$ ($k \neq 0$) arises from the outer involution $\rho \mapsto \bar{\rho}$ of the holonomy representation induced by complex conjugation in the invariant trace field of m006. Manifolds sharing the same invariant trace field and quaternion algebra have the same decay rate hierarchy.*

The rates c_k constitute a new family of geometric invariants of compact hyperbolic 3-manifolds with $H_1 \cong \mathbb{Z}/5$. Extending the census across the OrientableClosedCensus for other manifolds with $H_1 = \mathbb{Z}/p$ (p prime) would test whether the rate hierarchy depends on p , on arithmeticity, or on the specific quaternion algebra.

5 Conclusions

The homology-resolved spectral floor $\phi_{\text{floor}}^{(k)}(L)$ reveals a class-dependent 4:2:1 hierarchy in the loxodromic twist spectra of two compact hyperbolic 3-manifolds with $H_1 = \mathbb{Z}/5$. Exponential

decay with paired rates $c_1 = c_4 \approx 0.985$, $c_2 = c_3 \approx 0.401$, $c_0 \approx 0.750$ is confirmed by AIC model selection over $L = 1, \dots, 12$. The hierarchy persists in both manifolds despite their differing arithmetic invariants, implicating the $\mathbb{Z}/5$ torsion structure as the origin. A conjectured arithmetic explanation via complex conjugation in the invariant trace field is proposed for future verification via the Snap package.

Data Availability

Complete census files and scripts at <https://github.com/drmlgentry/hyperbolic-flavor-scan>.

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References

- [1] D. Gabai, R. Meyerhoff, P. Milley, J. Amer. Math. Soc. **22**, 1157 (2009).
- [2] C. Maclachlan, A. W. Reid, *The Arithmetic of Hyperbolic 3-Manifolds*, Springer (2003).
- [3] M. L. Gentry, “Hyperbolic Flavor Geometry,” SSRN 6775158 (2026); PTEP Paper No. 2605-035.
- [4] M. L. Gentry, “Quark Mixing from Hyperbolic Geometry,” SSRN 6583550 (2026).
- [5] M. L. Gentry, “Lepton Mixing from Borel Structure of Hyperbolic Holonomy,” SSRN 6583553 (2026); Phys. Lett. B PLB-D-26-01448.
- [6] A. Balog et al., “Ambient Prime Geodesic Theorems on Hyperbolic 3-Manifolds,” Int. Math. Res. Not. **2023**(1), 588–606.
- [7] G. A. Margulis, Funct. Anal. Appl. **3**, 335 (1969).
- [8] T. Sunada, Ann. Math. **121**, 169 (1985).
- [9] T. Chinburg, E. Friedman, K. N. Jones, A. W. Reid, Ann. Scuola Norm. Sup. Pisa **30**, 1 (2001).
- [10] M. Culler, N. M. Dunfield, M. Goerner, J. R. Weeks, SnapPy, <http://snappy.computop.org>.
- [11] D. Coulson, O. Goodman, C. Hodgson, W. Neumann, Experimental Math. **9**, 127 (2000).